

Max Hamilton

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Education

University of Massachusetts, Amherst

Sep 2022 - Dec 2026 (Expected)

Ph.D. in Computer Science

Advisor: Subhansu Maji

University of Michigan, Ann Arbor

Sep 2020 - May 2022

M.S.E. in Computer Science and Engineering

Cumulative GPA: 4.00/4.00

University of Michigan, Ann Arbor

Sep 2016 - May 2020

B.S.E. in Computer Science and Engineering

Minor in Mathematics

Cumulative GPA: 4.00/4.00

Research Experience

Computer Vision Lab, Graduate Research Assistant, Amherst, MA

Sep 2022 - Present

Advisor: Prof. Subhansu Maji

- Neurips 2025 paper: Introduced *Active Measurement*, a human-in-the-loop estimation framework combining importance sampling and Monte Carlo estimation to produce statistically unbiased, high-precision aggregate measurements from unlabeled data. Derived novel estimators, weighting schemes, and confidence intervals, demonstrating significantly lower error and reduced human labeling cost vs. baseline methods.
- Published first author paper at Neurips 2024 on zero-shot species range estimation using observational data and species text descriptions
- Designed and implemented a scalable text processing pipeline over 37k Wikipedia articles, generating section-level embeddings that were integrated into a multimodal training system to learn a text-to-range-map model.

Dolby Laboratories, PhD Research Intern, Sunnyvale, CA

June 2024 – Sep 2024

Supervisor: Jaclyn Pytlarz

- Researched and implemented efficient fine-tuning strategies for diffusion models on geometric vision tasks (depth and surface normal estimation)
- Matched state-of-the-art relative depth performance while reducing inference cost by 8×, enabling faster deployment in resource-constrained environments

Owens Research Group, Graduate Research Assistant, Ann Arbor, MI

Apr 2021 - Sep 2022

Advisor: Prof. Andrew Owens

- Developed correspondence-wise loss functions to achieve state-of-the-art results in video next frame prediction tasks
- Incorporated optical flow methods into loss function to compute correspondences between image predictions and ground truth
- Coauthored paper accepted to CVPR 2022

Fouhey AI Lab, Undergraduate Research Assistant, Ann Arbor, MI

May 2019 - Apr 2021

Advisor: Prof. David Fouhey

- Led project to perform weather pattern localization using machine learning in PyTorch.
- Built a large-scale dataset of 10M+ geo-tagged sky images to enable supervision at scale
- Explored self-supervised and implicit representations for image-based geo-localization

Vision and Learning Lab, Undergraduate Research Assistant, Ann Arbor, MI
 Advisor: Prof. Jia Deng

May 2018 - Jan 2019

- Implemented a two-stage optimization pipeline in C++ to reconstruct 3D meshes from annotated 2D images
- Applied numerical optimization techniques to solve constrained curvature minimization for surface normal estimation
- Built a custom web crawler to collect 140,000+ real-world object images and extract camera pose from EXIF metadata

Publications

Active Measurement of Two Point Correlations

Max Hamilton, Subhansu Maji, Daniel Sheldon

In Submission, 2026.

Active Measurement: Efficient Estimation at Scale

Max Hamilton*, Jinlin Lai*, Wenlong Zhao, Subhansu Maji, Daniel Sheldon

Advances in Neural Information Processing Systems (NeurIPS), 2025.

Feedforward Few-shot Species Range Estimation

Christian Lange, **Max Hamilton**, Elijah Cole, Alexander Shepard, Samuel Heinrich, Angela Zhu, Subhansu Maji, Grant Van Horn, and Oisin Mac Aodha

Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.

Combining Observational Data and Language for Species Range Estimation

Max Hamilton, Christian Lange, Elijah Cole, Alexander Shepard, Samuel Heinrich, Oisin Mac Aodha, Grant Van Horn, and Subhansu Maji

Advances in Neural Information Processing Systems (NeurIPS), 2024.

Comparing Correspondences: Video Prediction with Correspondence-wise Losses

Daniel Geng, **Max Hamilton**, and Andrew Owens

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

OASIS: A Large-Scale Dataset for Single Image 3D in the Wild

Weifeng Chen, Shengyi Qian, David Fan, Noriyuki Kojima, **Max Hamilton**, Jia Deng

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

Teaching Experience

University of Massachusetts Amherst, Teaching Assistant

CompSci 670: Computer Vision

Spring 2024

CompSci 682: Neural Networks

Fall 2023

University of Michigan, Graduate Student Instructor

EECS 542: Advanced Topics in Computer Vision

Winter 2022

Honors & Awards

Edward Riseman and Allen Hanson Scholarship, University of Massachusetts, Amherst

Apr 2023

Christopher P. Moeller Scholarship Award, University of Michigan

Sep 2021

Awarded to outstanding graduate students studying computer science or mathematics.

EECS Scholar, University of Michigan

2019 - 2020

Tau Beta Pi Honor Society, University of Michigan

Apr 2018

Dean's List, University of Michigan

2016 - 2020

Professional Activities

Mentor, Early Research Scholars Program, University of Massachusetts, Amherst 2024 – Present
Program committee, New England Computer Vision (NECV) Workshop 2025
Reviewer: ICCV 2023, WACV 2024, ECCV 2024, WACV 2025, NeurIPS 2025, CVPR 2026

Skills

Programming Languages: Python, SQL, C++, MATLAB
Distributed Computing: Amazon Web Services, Google Cloud Platform, Slurm
Machine learning: PyTorch, PyTorch-lightning, NumPy, Huggingface, TensorBoard, Optuna